

2-Stage Miller- Compensated CMOS Operational Amplifier

**PVT-Verified Design, Stability Analysis, and Performance
Characterization**

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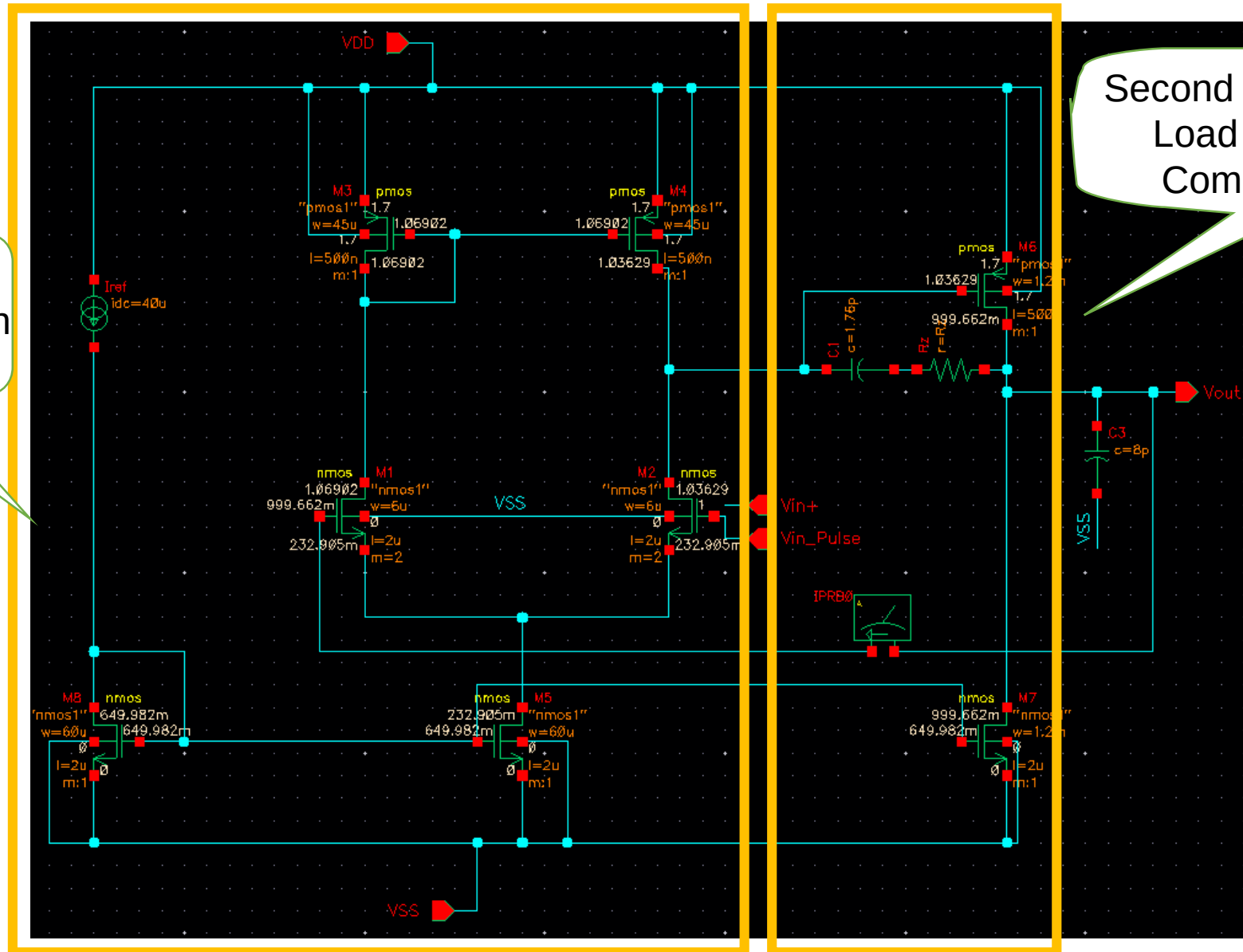
Target Specifications

(using gpdk180)

Requirement	Value
Base Length (L)	2um
Supply Voltage (VDD / VSS)	1.8V / 0V
Load Capacitance (CL)	8pF
Open-Loop Gain (Av)	>2500 V/V (>68dB)
Unity Gain Bandwidth (GBW)	>= 8MHz
Phase Margin (PM)	>= 60°
Slew Rate (SR)	> 15 V/us
Input Common-Mode Range (ICMR)	1.0V to 1.6V
Output Voltage Swing (OVS)	0.4V to 1.2V
Max Power Dissipation (P _{diss})	<= 2.4 mW

Schematic Architecture

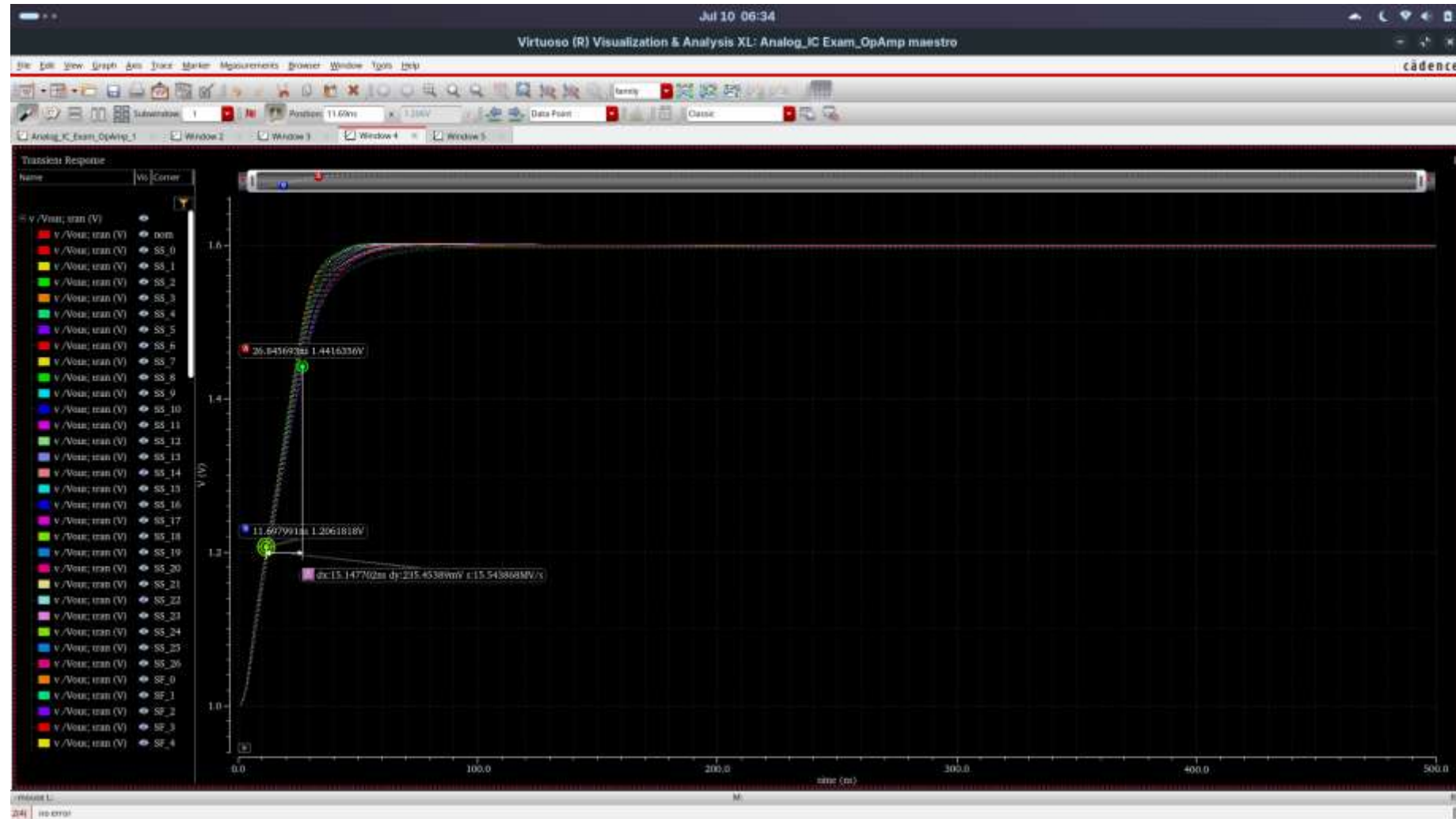
First Stage:
Differential Pair with
Current Mirrors



Transistor Sizing Chart

Device	Function	W/L Ratio
M1, M2	NMOS Differential Pair	12u/2u
M3, M4	PMOS Active Load	45u/0.5u
M5, M8	NMOS Tail Current / Bias	60u/2u
M6	PMOS Second Stage	1.2m/0.5u
M7	NMOS Second Stage Bias	1.2m/2u
Cc	Miller Capacitor	1.76 pF
Rz	Nulling Resistor	2k ohms

Simulation & Verification Results: Frequency and Transient Response



Slew Rate =
15.52V/us

Simulation & Verification Results: AC Gain and Phase Margin



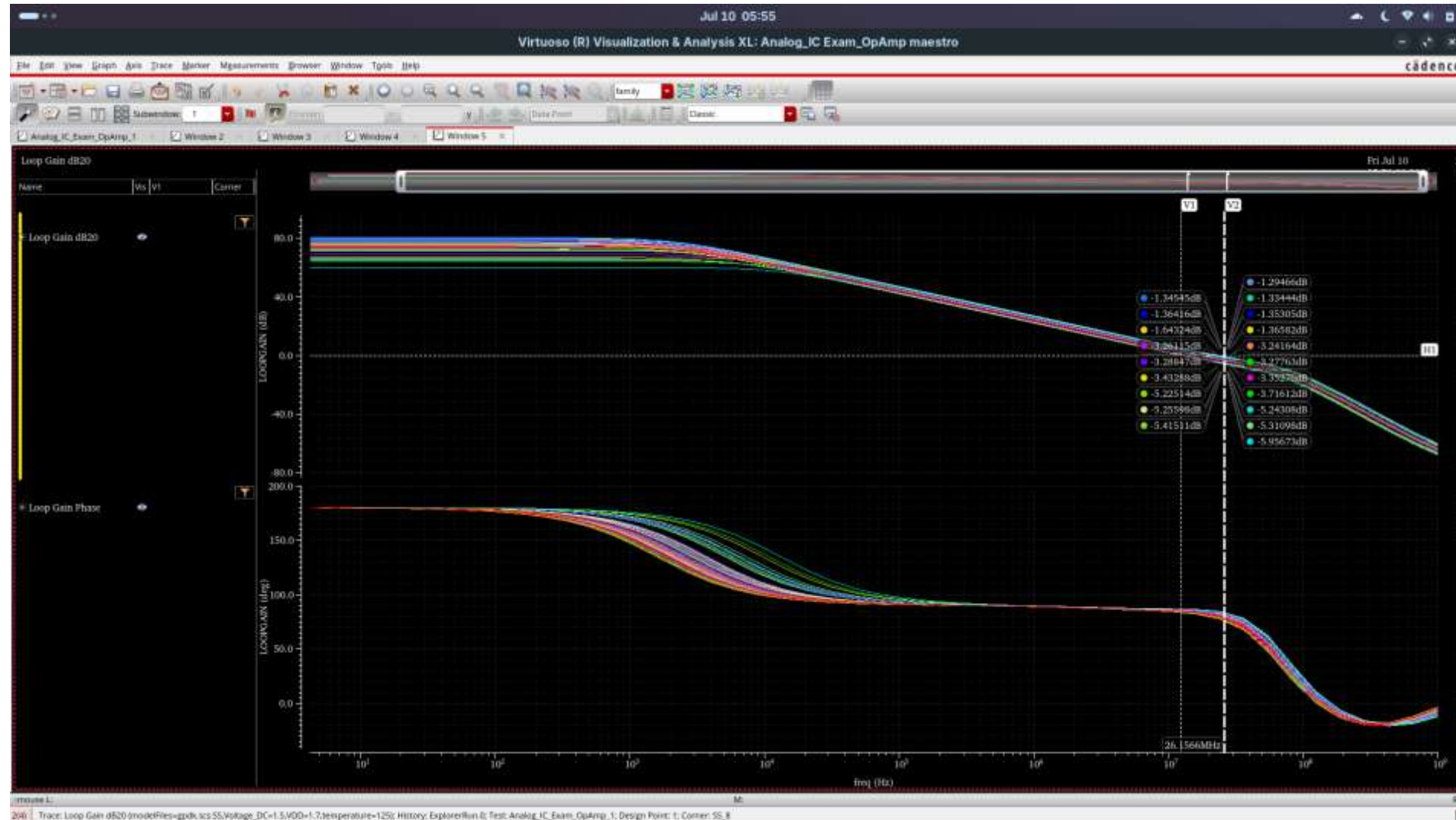
Highest Gain:
80dB

Lowest Gain:
60dB

Highest BW:
26.16 MHz

Lowest BW:
12.49 MHz

Simulation & Verification Results: AC Gain and Phase Margin



Highest Gain:
80dB

Lowest Gain:
60dB

Highest BW:
26.16 MHz

Lowest BW:
12.49 MHz

Simulation & Verification Results: AC Gain and Phase Margin



Highest PM:
86 degrees

Lowest PM:
83 degrees

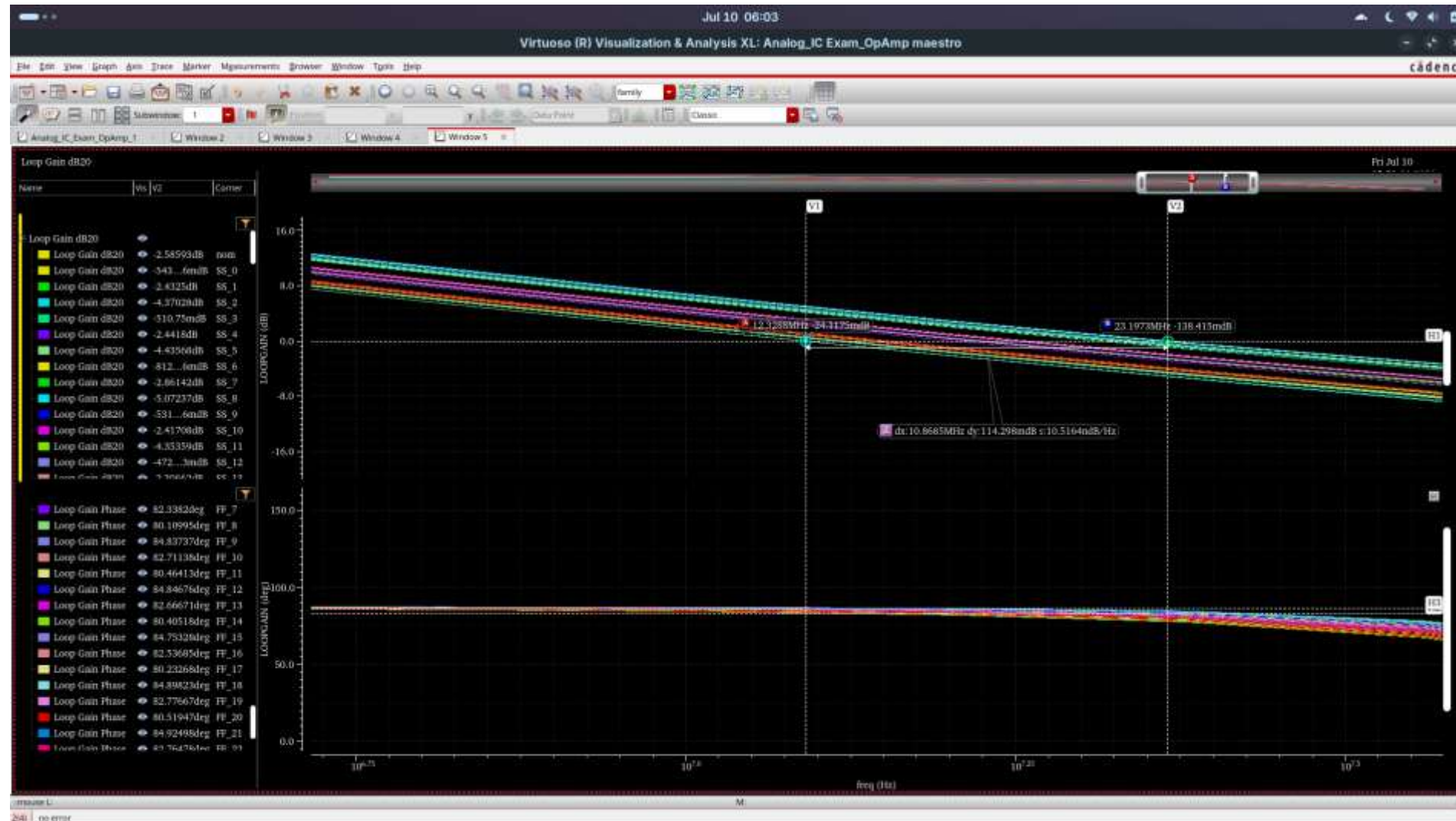
Simulation & Verification Results: AC Gain and Phase Margin



Highest PM:
86 degrees

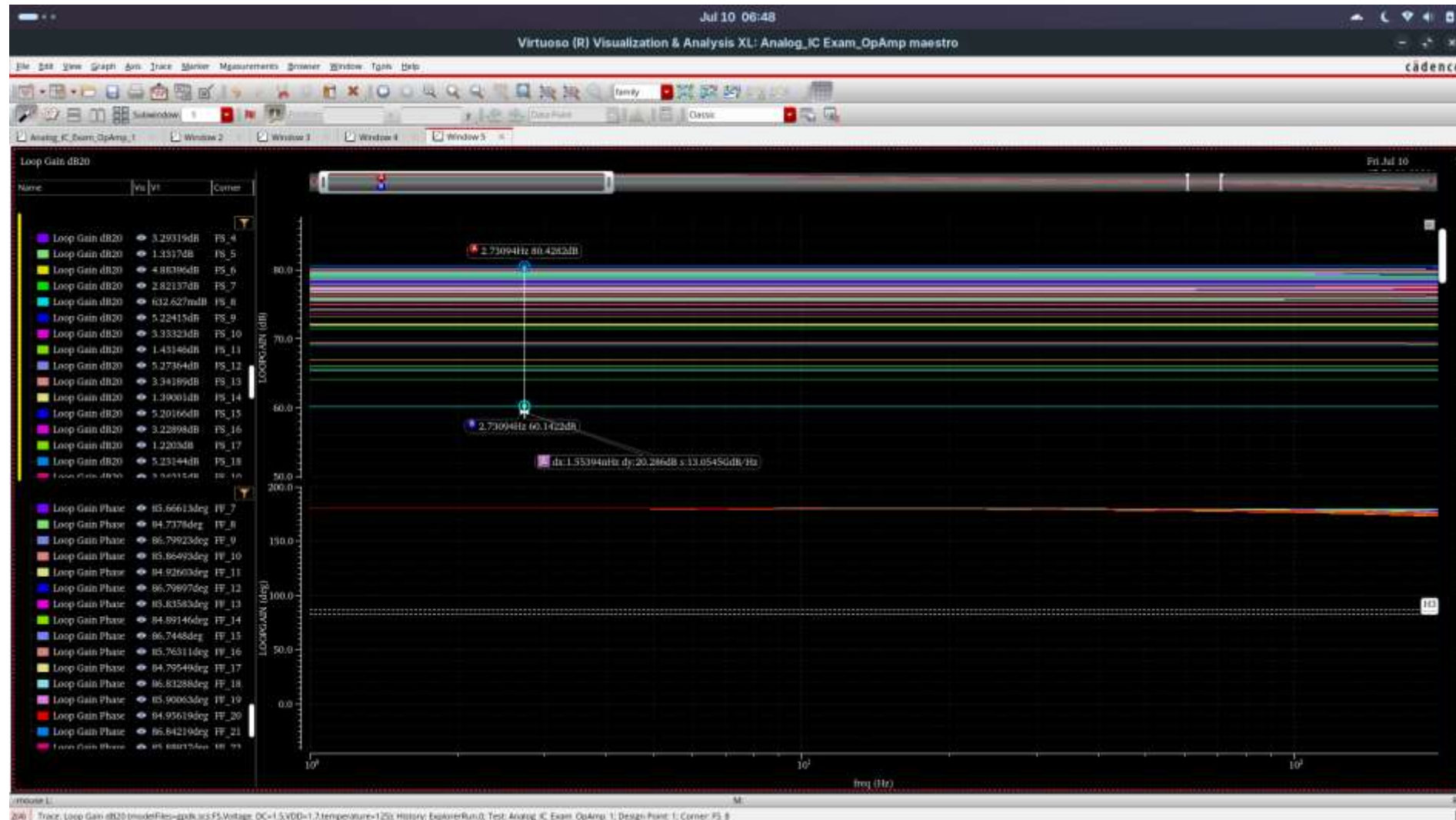
Lowest PM:
83 degrees

Simulation & Verification Results: AC Gain and Phase Margin



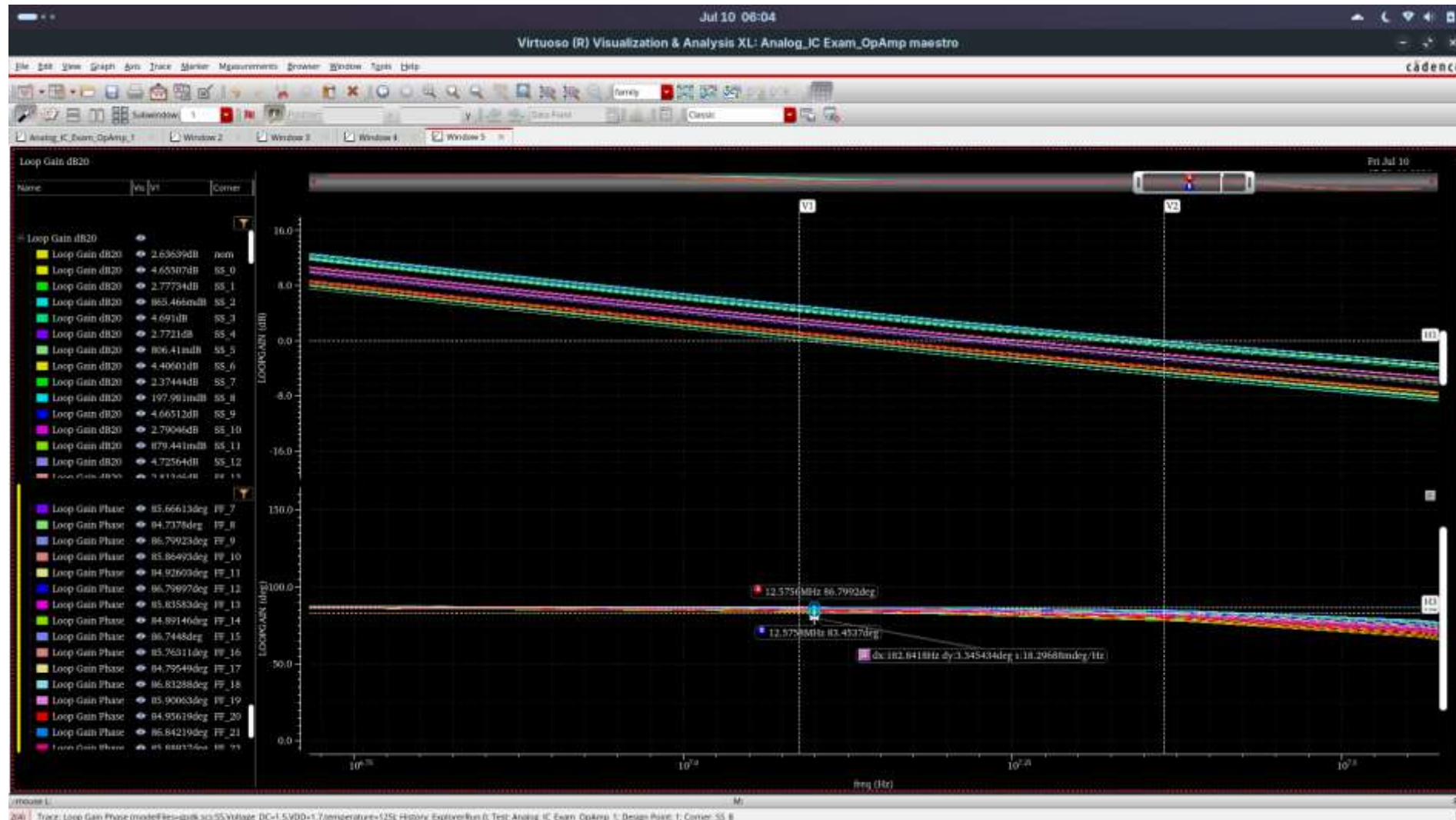
Highest -
Lowest BW:
10.86 MHz

Simulation & Verification Results: AC Gain and Phase Margin



Highest -
Lowest Gain:
20.28 dB

Simulation & Verification Results: AC Gain and Phase Margin



Highest -
Lowest PM
at Lower BW
point:

3.34 degrees

Simulation & Verification Results: AC Gain and Phase Margin



Highest -
Lowest PM
at Higher BW
point:

6.87 degrees

Performance Specification Sheet: PVT Corner Validation, across SS, SF, FS, FF corners.

Parameter	Minimum	Nominal	Maximum	Remarks
Supply Voltage (VDD)	1.7V	1.8V	1.89V	gpdk180 process
Temperature	-40 °C	27 °C	125 °C	Industrial Range
Functional ICMR	1.0V	1.3V	1.5V	Unity-Gain config
Phase Margin (PM)	83.4°	84.1°	85.1°	All PVT Corners
Unity Gain Bandwidth (GBW)	12.2 MHz	16.4 MHz	23.2 MHz	Worst Case: SS / 125 °C
Gain Margin (GM)	24.1 dB	29.2 dB	29.6 dB	All PVT Corners

Performance Specification Sheet: Target vs. Achieved Verification

Parameter	Target Specification	Achieved Result (Worst Case)
Open-Loop Gain (A_v)	$>2500 \text{ V/V}$ ($>68\text{dB}$)	$>15,000 \text{ V/V}$ ($\sim 84\text{dB}$)
Unity Gain Bandwidth (GBW)	$\geq 8\text{MHz}$	12.2MHz
Phase Margin (PM)	$\geq 60^\circ$	83.4°
Slew Rate (SR)	$> 15 \text{ V/us}$	$\sim 15.9 \text{ V/us}$
Input Common-Mode Range (ICMR)	1.0V to 1.6V	1.0V to 1.5V
Output Voltage Swing (OVS)	0.4V to 1.2V	0.4V to 1.2V
Max Power Dissipation (P_{diss})	$\leq 2.4 \text{ mW}$	0.26mW

- Gain & Stability Over-Performance: The amplifier significantly exceeded the open-loop gain and phase margin targets.
- ICMR Architectural Compromise: The original requirement of a rail-to-rail 1.8V ICMR was physically incompatible with the 5-transistor PMOS-load architecture at a 1.7V minimum supply corner.

Boundary Conditions & Physical Constraints

- Thermal Transconductance Degradation: Carrier mobility drops significantly at the upper thermal boundary (125°C). While the circuit remains fully stable, the Unity Gain Bandwidth experiences a nominal reduction from ~23 MHz (Fast corners, -40°C) down to a worst-case 12.2 MHz.
- Strict 1.5V Input Common-Mode Ceiling: The amplifier's architecture relies on a PMOS active load clamping the drain nodes of the input differential pair. Exceeding a 1.5V input signal on a minimum 1.7V supply severely starves the differential pair of VDS headroom, driving them into the triode region and causing a catastrophic collapse of open-loop gain.

Practical Applications & Deployment

- Precision Analog Front-End (AFE) Buffer: The exceptionally high phase margin ($>83^\circ$ across all corners) ensures a completely critically damped step response. This makes the op-amp ideal for precision closed-loop amplification (e.g., non-inverting topologies scaling 10kHz AC signals) with zero high-frequency ringing or overshoot.
- Mid-Rail Sensor Signal Conditioning: Highly capable of buffering low-frequency analog sensor outputs (such as color or temperature monitors) where the signal swings strictly between 1.0V and 1.5V.
- On-Chip Reference Distribution: The stable capacitive driving capabilities make it suitable for acting as a localized buffer for bandgap reference voltages, safely isolating the reference generator from noisy sub-blocks across an IC.